Ground Zero: The Chernobyl Response Group

Thomas Jefferson Model United Nations Conference

TechMUN XXXIII

Middle School Specialized Agency

Chairs: Siri Lingutla and Meerub Rana

Thomas Jefferson High School for Science and Technology

Disaster Background:

In the early hours of April 26, 1986, occurred an explosion and catastrophic meltdown of the Chernobyl Nuclear Power Plant's reactor unit four in what was a nighttime safety test that led to the leaking of millions of curies of radioactive material into Ukraine, Europe, and the eastern USSR. The accidents were a result of the failings of technological design, operational errors, and lack of a safe culture. The Soviet Union's initial inadequate evacuation planning and widespread release of radioactivity from the Chernobyl Nuclear Power Station created a humanitarian, environmental, and political disaster that will remain for generations.

Topic A: The 48-Hour Threshold: Containment & The Thermal Threat

Background:

The explosion at the Chernobyl Nuclear Power Plant occurred on April 26, 1986, in Pripyat, Ukraine, when a reactor exploded. This explosion is attributed to human error, flawed design, and operating problems with reactor No. 4 (RBMK-1000). As a result of this error, a steam explosion occurred that blew the 1,000-ton biological shield (or lid) off the reactor, exposing the nuclear core to atmospheric air. The use of large quantities of graphite as a neutron moderator in the reactor resulted in the combustion of graphite when air was introduced. This resulted in an extraordinary, open-air nuclear fire resulting in large quantities of radioactive materials being pumped into the upper atmosphere through the chimney of the reactor.

Aside from the immediate destruction caused by the atmospheric contamination, an additional danger exists, a much larger danger. Underneath the burning core of the reactor are large pools of water (called "bubbler pools") that were designed to hold several million liters of water. These bubbles were placed there to condense the steam if a pipe that was bringing coolant to the Reactor Core broke. With the core of the reactor now melted and exposed to air, molten lava is forming underneath the core, with a temperature of 2500 degrees Celsius. If the lava from the core burns through the rest of the concrete floor and falls into the bubbler pools, there will be an explosion due to the rapid creation of steam from water. The explosion of the steam is predicted to produce a force equivalent to 4 megatons of TNT. An explosion of that size would destroy the existing three nuclear reactors located at Chernobyl instantly and cause enough lethal radioactive materials to be released into the environment to make much of Eastern Europe uninhabitable for many thousands of years.

Current Situation:

After the explosion, the committee convened within a few hours after the incident. The workers in the plant and local residents don't know how life-threatening this situation is and the graphite fire is out of control. When attempting to fight the fire, firefighters found that using water was both ineffective and increased the amount of radioactive steam, when water comes in contact with the graphite fire, it vaporizes instantly and adds more radioactive steam into the area. First responders, including firefighters from local areas and the plant operators have moved into the area to attempt to bring the fire under control, however they are experiencing fatal doses of radiation attempting to do this.

Under the reactor, there are large pools of water called bubbler pools. Right now, the melted nuclear fuel inside the reactor is extremely hot and is burning downward. The machines that measure temperature were destroyed in the explosion, so no one knows exactly how hot the core is or how fast it is melting. If the melted fuel burns through the floor and falls into the water below, the water could instantly turn into steam and cause another massive explosion. The committee must focus on two urgent actions: First, they must put out the graphite fire from above so it stops sending radioactive smoke into the air. Second, they must drain the water from underneath the reactor so that if the melted fuel reaches the bottom, it will not cause another explosion.

The evacuation protocols for Pripyat (50,000 population) have not begun to take place so the civilian population is still exposed to potentially lethal amounts of radioactive Iodine-131 and Cesium-137. The members of the committee recognize that every hour that passes allows for two things: 1) Options decrease and 2) The death rate from the exposure to high levels of radiation continues to increase.

Past Actions:

Firefighters at the Chernobyl nuclear plant complied with standard firefighting protocol (for a run-of-the-mill building fire) immediately following the explosion however this was not a standard fire, it was much worse, the nuclear reactor made it almost impossible for safe rescue efforts. Initially, officials who were in charge of the plant insisted to government authorities in Moscow that no explosion took place, and it was only a roof fire. Firefighting crews were dispatched to extinguish the fire without the proper protective gear as they were misled about the nature of the fire and some very high-risk areas where the firefighting crews had to work were subjected to incredibly high levels of radiation thus causing many of them acute radiation syndrome.

The Soviet Union at that time was organized by a few members of the ruling party who made most of the decisions without the input of other officials and information regarding any major incidents would not be made available to the public due to the government being concerned that if the public knew there would be a panic. This incident was too catastrophic for it to remain a secret, and the government was not prepared nor did they have the proper resources to control an outrage from the public.

Possible Solutions:

One option for containing the shattered reactor is to have military helicopters fly over it and drop large amounts of sand and clay to create a barrier that would limit the amount of radioactive smoke being produced. This would pose a significant risk to the pilots, since they will be flying over extremely high levels of radiation. Another key to preventing a second explosion is for trained engineers to enter the flooded basement under the reactor and open a series of gates that let water drain from the bubbler pool. This task also represents a substantial threat to the engineers as they will be entering an area with extremely high radiation levels.

The committee must also create an effective plan to assist all those individuals who have already been affected by radiation. Medical personnel will need to secure supplies as well as ensure that medical facilities are prepared to treat radiation-exposed individuals. Additionally, local leaders must develop and implement an orderly and safe plan to remove residents from within the zone of danger in order to eliminate or reduce their exposure. If the military, the scientific community, and the medical community collaborate closely and work expeditiously, they may prevent a second explosion from occurring and position themselves to begin planning for medium- and long-term cleanup and recovery.

Guiding Questions

1. What specific logistics will be required to move the required materials (boron, lead, sand) and military aircraft to extinguish the reactor fire?
2. How will the committee develop and carry out the high-risk, manual removal of water from bubbler pools and who will be determined for this operation?
3. What is the timeline and perimeter for the evacuation of people from Pripyat, and what method will be used to communicate to the public about these plans while not inciting mass panic?
4. How should medical supplies and advanced medical treatments be distributed for the first responders who have developed Acute Radiation Syndrome?
5. How can the committee obtain accurate radiation measurements and diagnostic information from the core given that standard instruments have been destroyed or exposed to lethal levels of radiation?
6. If helicopter containment drops fail, what other operational methods are available to smother the reactor fire?

Topic B: The Groundwater Threat

Background:

At the Chornobyl Nuclear Power Plant, the highly radioactive and extremely hot melted nuclear fuel from the reactor has combined with materials such as concrete, metal, and various items to form a substance known as corium. Although the melted corium continues to burn downward, the fire was initially reduced with the introduction of sand, clay, and lead, it remains at a temperature of over 2000 degrees Celsius and will continue to slowly melt through the thick concrete floor of the building.

This phenomenon has typically been referred to by scientists as the "China Syndrome". It's used to describe how an object that has melted continues to melt downward and will then create a more serious issue underground. The area below the power station contains groundwater, referred to as aquifers, which consists of underground water that connects through the Pripyat River to the Dneiper River and has been used as a source of drinkable water for over 30 million people. Should the melted corium continue to create a hole through the concrete and come into contact with groundwater, the drinking supply for many people could become contaminated. This scenario has the potential of harming millions of human beings and also severely impacting all of the crops and farms located in the region.

Current Situation:

Cooling the reactor from the top doesn't appear to be effective. The amount of sand, clay, and other materials added to the reactor has added weight to a building that has already experienced significant damage. The radiation levels inside the facility are so high that they can kill someone quickly, meaning that any attempt to stop the corium will be done with extreme risk.

There is disagreement among scientists and government officials throughout the former Soviet Union as to how long until the melted material will penetrate the floor of the reactor and hit the body of water underneath the plant. The government must take urgent and large-scale action, which could mean moving large amounts of money, personnel and equipment from all areas of the country in order to address a single problem. As events continue to change rapidly, the workers who are involved must look out for themselves, as well as work on stopping a potential catastrophic event that could make the country's water supply unfit for use.

Past Actions:

The Soviet Union had a very large and organized mining industry. In the past, miners were often used for big government building projects, like tunnels and underground construction. But this was the first time miners were asked to help with an underground nuclear emergency. At the beginning of the disaster at the Chornobyl Nuclear Power Plant, workers focused on putting out the fire from above the reactor. They dropped huge amounts of sand and other materials onto it. This helped reduce the fire, but it also created a heavy cap on top of the reactor. That weight is now pushing the melted corium downward.

Soviet leaders frequently addressed large industrial problems by deploying massive numbers of workers. Early in the disaster, soldiers entered highly radioactive zones without adequate protection to assess the damage. Many became gravely ill or died from radiation. No machines could withstand the intense radiation inside the plant. Technology at the time could not survive the powerful gamma rays long enough to drill or measure conditions. As a result, the government relied almost entirely on human labor and basic tools, despite the extreme danger.

Possible Solutions:

In order to prevent contaminated melted corium from leaking into underground water sources, delegates could consider starting a large excavation project beneath reactor #4 at the Chernobyl Nuclear Power Plant site. The delegates would enlist experienced mining workers from the nearby areas of Tula or Donbas as well as other experienced miners, to dig a tunnel underneath the damaged reactor so that there would be an underground area directly below it. After completing the construction of the underground area, workers would need to construct an insulated large cooling pad surrounding it. First, they would pump extremely cold liquid nitrogen into the ground to freeze the soil to prevent heating being transferred out of the area. Once the ground has been frozen the workers will then pour a thick layer of reinforced concrete on top of ground level to create a strong barrier to prevent further corium from making its way down and into the underground water below.

The work required to accomplish this task would be very hazardous to the workers. They would be required to work in extreme heat with very little oxygen, and extreme radiation levels. Delegates will need to make and implement directives to minimize worker risks and ensure the tunnel is completed before explosives are used to blow apart the remaining structure of the damaged building above.

Guiding Questions

1. How does the committee intend to rapidly source, transport, and accommodate the thousands of skilled miners needed for the underground excavation?
2. What levels of safety and/or radioactive exposure will be observed for miners working directly under the melting core?

3. How will the large amounts of liquid nitrogen and special concrete needed to create the heat exchanger be provided and delivered to the Exclusion Zone?
4. What type of contingency plans will be established to prevent a collapse of the reactor's foundation if the excavation timetables speed up?
5. How will the state make restitution or provide long-term healthcare to any of the mining workforce who were subject to these extreme conditions?
6. If the underground cooling pad fails to stop the corium, what is the Soviet Union's plan to guard against total contamination of the Dnieper River basin?

Works Cited

1. "Chernobyl Disaster | Causes, Effects, Deaths, Videos, Location, & Facts." Britannica, Encyclopaedia Britannica, https://www.britannica.com/event/Chernobyl-disaster
2. Nuclear Regulatory Commission. Backgrounder on Chernobyl Nuclear Power Plant Accident.U.S. Nuclear Regulatory Commission, https://www.nrc.gov/reading-rm/doc-collections/fact-sheets/chernobyl-bg
3. United Nations Scientific Committee on the Effects of Atomic Radiation. The Chornobyl Accident. UNSCEAR, https://www.unscear.org/unscear/en/areas-of-work/chernobyl.html
4. World Nuclear Association. Chernobyl Accident 1986. World Nuclear Association, 17 Feb.2025,https://world-nuclear.org/information-library/safety-and-security/safety-of-plants/chernobyl-accident

5. Seventeen Moments in Soviet History Press Team. "Economic and Social Consequences of the Chernobyl Accident." *Seventeen Moments in Soviet History*, 3 Apr. 2020, http://soviethistory.msu.edu/1985-2/meltdown-in-chernobyl/meltdown-in-chernobyl-texts/economic-and-social-consequences-of-the-chernobyl-accident/.
6. World Health Organization. *Health Effects of the Chernobyl Accident and Special Health Care Programmes*. WHO, 2006, https://www.who.int/publications/i/item/9789241594174.
7. International Atomic Energy Agency. *The Chernobyl Accident: Updating of INSAG-1*. INSAG-7, IAEA, 1992, https://www-pub.iaea.org/MTCD/publications/PDF/Pub913e_web.pdf.

Dossier:

1. **Mikhail Gorbachev** – General Secretary of the CPSU; served as a leading decision-maker within the party, guiding committee discussions and promoting *glasnost* reforms while working to preserve overall party stability.
2. **Nikolai Ryzhkov** – Soviet Premier; coordinated national emergency management and allocated state funds for the massive cleanup effort.
3. **Yegor Ligachev** – Senior Politburo member; opposed transparency and enforced strict party discipline to suppress alarmist reports.
4. **Viktor Chebrikov** – KGB Chairman; controlled information flow, monitored public reaction, and severed communications to prevent leaks.
5. **Eduard Shevardnadze** – Soviet Foreign Minister; managed international diplomatic fallout and negotiated foreign medical aid.
6. **Volodymyr Shcherbytsky** – First Secretary of Soviet Ukraine; governed the disaster zone and enforced public normalcy despite radiation risks.
7. **Boris Shcherbina** – Deputy Chairman; led the government commission on the ground and authorized immediate containment measures.
8. **Valery Legasov** – Lead scientist; investigated the reactor flaws and advocated for technical truth against political pressure.
9. **Anatoly Aleksandrov** – President of the Soviet Academy of Sciences; designed the RBMK reactor and defended its safety record to protect his legacy.
10. **Dr. Leonid Ilyin** – Chief radiologist of the USSR; established public health protocols and determined acceptable radiation exposure limits.

11. **Vasily Nesterenko** – Belarusian nuclear physicist; operated independent monitoring networks and demanded protection for northern regions.
12. **Vladimir Pikalov** – Commander of Chemical Troops; led initial reconnaissance missions and secured accurate radiation readings from the core.
13. **Nikolai Tarakanov** – Commander of Liquidators; mobilized reservists for debris removal and managed high-risk roof clearing operations.
14. **Yuri Samoilenko** – Deputy Chief Engineer; procured remote-controlled machinery and devised engineering solutions for decontamination.
15. **Alexei Ananenko** – Mechanical Engineer; volunteered to drain the bubbler pools beneath the reactor to prevent a thermal explosion.
16. **Viktor Bryukhanov** – Plant Director; responsible for overall operations and delayed reporting the full extent of the accident.
17. **Nikolai Fomin** – Chief Engineer; approved the fatal safety test and attempted to blame shift operators for the explosion.
18. **Anatoly Dyatlov** – Deputy Chief Engineer; supervised the failed safety test and denied the possibility of core destruction.
19. **Alla Yaroshinskaya** – Investigative journalist; uncovered government cover-ups regarding civilian radiation exposure and secret protocols.
20. **Yelena Bonner** – Human-rights activist; leveraged Western media connections to highlight the systemic failures of the Soviet regime.
21. **Olha Ivanivna** – Pripyat City Council representative; advocated for the immediate evacuation and medical needs of displaced residents.

22. **Gennadi Gerasimov** – Foreign Ministry spokesman; managed the global media narrative and deflected international inquiries.
23. **Dr. Robert Gale** – American hematologist; coordinated Western medical relief and treated acute radiation victims in Moscow.
24. **Hans Blix** – Director General of the IAEA; demanded international inspections and influenced global nuclear safety standards.
25. **Margaret Thatcher** – UK Prime Minister; utilized intelligence on the disaster to pressure the USSR in Cold War negotiations.